Project Initialization and Planning Phase

Date	15 July 2026
Team ID	
Project Title	A Comprehensive Measure of Well-being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report
The proposal report aims to develop a Machine Learning-based Human Development Index (HDI) Prediction System that predicts a country's HDI using key socio-economic indicators. It addresses the challenges of manual HDI calculation by providing a fast, reliable, and easy-to-use web-based prediction tool.

Project Overview	
Objective	The primary objective of this project is to develop a Machine Learning-based Human Development Index (HDI) Prediction System that accurately predicts a country's HDI using key socio-economic indicators such as Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI) per Capita. The system aims to provide quick, reliable, and data-driven predictions through an easy-to-use web application.
Scope	The project focuses on analyzing HDI data, preprocessing the dataset, training a Linear Regression model, and deploying the prediction system using Flask. The application enables users to input socio-economic indicators and instantly receive the predicted HDI score along with its development category (Low, Medium, High, or Very High).
Problem Statement	
Description	Predicting the Human Development Index manually requires analyzing multiple socio-economic indicators and performing complex calculations, making the process time-consuming and prone to errors. Researchers, students, and policymakers often lack a simple tool for quick HDI estimation.
Impact	Providing an automated HDI prediction system improves efficiency, supports research and policy planning, reduces manual effort, and enables faster, data-driven decision-making.
Proposed Solution	
Approach	The proposed solution uses a Linear Regression Machine Learning model trained on the Human Development Index dataset. The model is integrated with a Flask web application, allowing users to enter socio-economic indicators and instantly receive the predicted HDI score and development category.
Key Features	● Machine Learning-based HDI prediction. ● User-friendly web interface using Flask. ● Instant prediction of HDI score. ● Classification into Low, Medium, High, or Very High development. ● Accurate predictions using a trained Linear Regression model. ● Simple and responsive interface for easy interaction.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Intel/AMD Processor (Dual-Core or above)
	Memory (RAM)	8 GB
	Storage	1 GB Free Disk Space
Software	Framework	Flask
	Libraries	Scikit-learn, Pandas, NumPy, Pickle
	Development Environment	Visual Studio Code, Jupyter Notebook
Data	Dataset	Human Development Index (HDI) Dataset (CSV Format)